

IOT BASED RAIN ALARM AND SHUTTER SYSTEM

PROJECT GUIDE:

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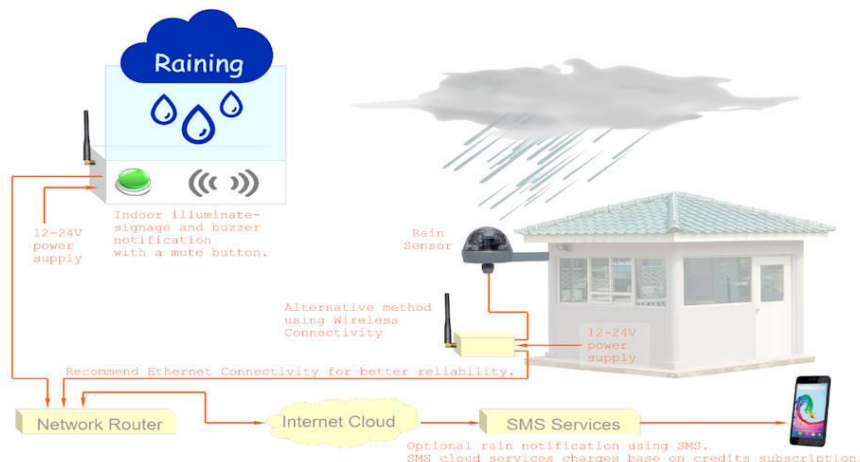


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Rain Detection & Alert Notification System



INTRODUCTION

In an increasingly interconnected world, the need for smart and efficient systems to manage our homes and environments has never been more apparent. One such necessity is the implementation of rain shutters and alarm systems.

These systems not only protect our homes from the elements but also provide a crucial layer of security.

However, their operation and management can often be cumbersome and time-consuming. This is where the Internet of Things (IoT) comes into play. IoT, with its ability to interconnect devices and automate tasks, can significantly ease the task of managing these systems.

OBJECTIVE

- ❑ The primary goal of this project is to create an IoT-based **RAIN ALARM AND SHUTTER** system that will simplify and automate the operation of rain shutters and alarm systems.
- ❑ It helps in improving efficiency and user-friendliness while also contributing to the advancement of smart home technology.
- ❑ It will also show outside data such as humidity and temperature levels during a rainstorm.

COMPONENTS USED:

The hardware consist of :

- ✓ Arduino UNO
- ✓ NodeMCU (ESP32)
- ✓ Rain sensor
- ✓ Temperature and Humidity sensor (DHT 11)
- ✓ Water droplets
- ✓ Breadboard
- ✓ Connecting wires
- ✓ 12V DC Battery

ARDUINO UNO

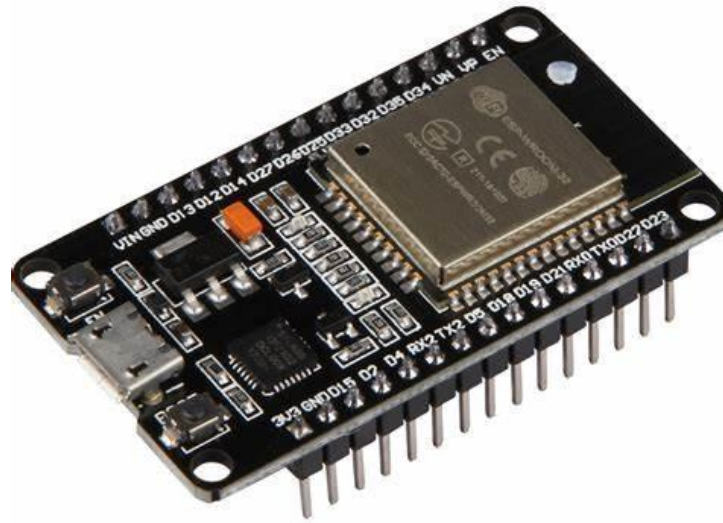
The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P microcontroller.

The board is equipped with sets of digital and analog input/output (I/O) pins that may be interfaced to various expansion boards and other circuits.



NodeMCU (ESP32)

The ESP32 is a series of low-cost, low-power system on a chip microcontrollers with integrated Wi-Fi and dual-mode Bluetooth. It's widely used in IoT projects due to its wireless connectivity options and its ability to run complex applications.



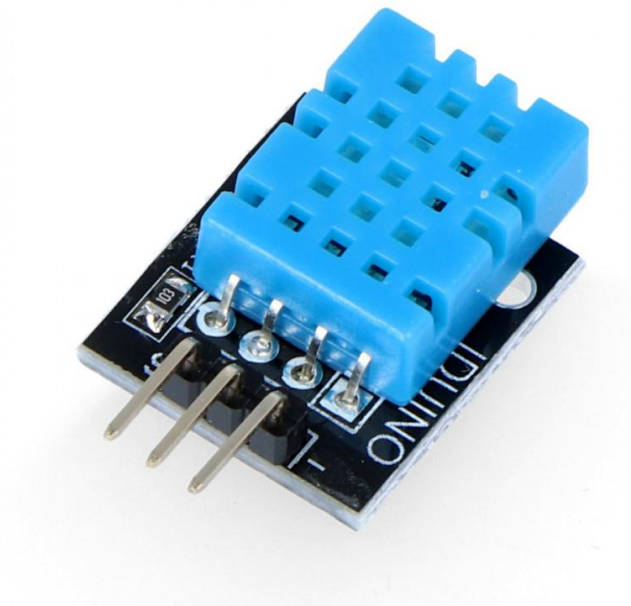
Rain Sensor

A rain sensor or rain switch is a switching device activated by rainfall. They are used in automatic wiper systems in cars or to shut windows automatically in smart homes.



Temperature and Humidity Sensor

Temperature and humidity sensors are devices that can measure the humidity and the temperature of the environment. They work by finding the amount of water vapor present in the air around the sensors .They can convert temperature and humidity into electrical signals that can be easily read



WORKING

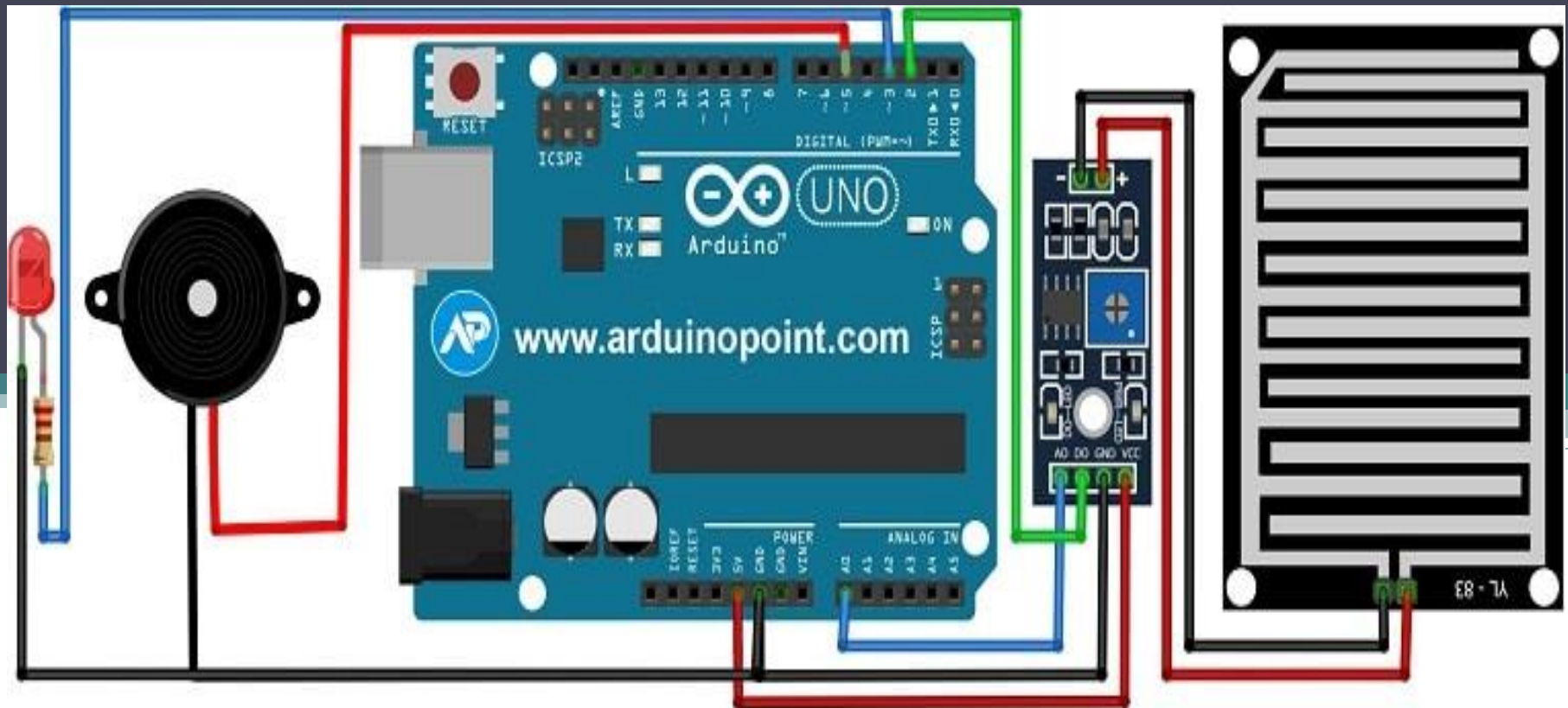
Here is how a rain alarm and shutter system can work using Arduino Uno and ESP32.

- The rain sensor detects rainfall and sends a signal to the Arduino Uno. The Arduino reads the analog output of the rain sensor and determines if it is raining based on a preset threshold.
- If rain is detected, the Arduino sends a signal to activate a buzzer to sound the alarm and it also activates the motors to open the shutters.
- The esp32 will send the outside data about temperature and humidity to the user's mobile phone using the platform called Thingspeak.
- Once the rain stops, the rain sensor will send a signal to the Arduino indicating dry conditions.
- The Arduino will then send a signal to deactivate the alarm and close the shutters.

This system uses the rain sensor to detect rain, the Arduino Uno to read the sensor data and determine the control signals, activate the alarm and shutters, and Wifi connectivity of the esp32 to send data on Thingspeak platform.

The components work together to automatically detect rain and activate the necessary actions, and alert the user about the events.

CIRCUIT DIAGRAM



ABOUT ThingSpeak

ThingSpeak is an IoT analytics platform service that allows you to aggregate, visualize, and analyze live data streams in the cloud. You can send data to ThingSpeak from your devices, create instant visualization of live data, and send alerts.



Humidity and Tempurature Monitoring

Channel ID: 2167814

Author: [mwa0000029991347](#)

Access: Private



Channel Stats

Created: [19 days ago](#)

Last entry: [13 days ago](#)

Entries: 13



Field 1 Chart



ThingSpeak.com

Field 2 Chart



ThingSpeak.com

Tempurature



31

13 days ago

Humidity



36

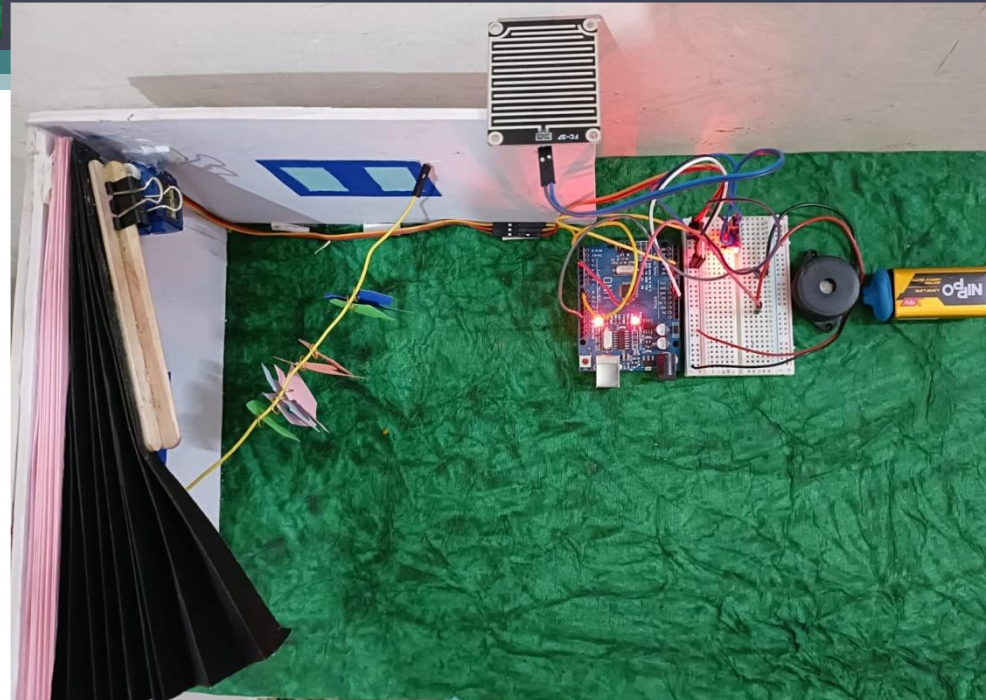
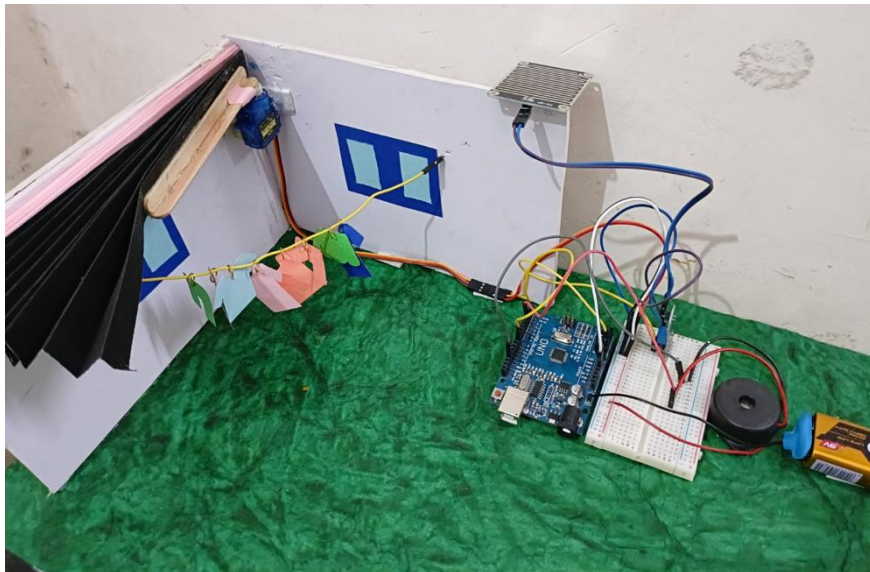
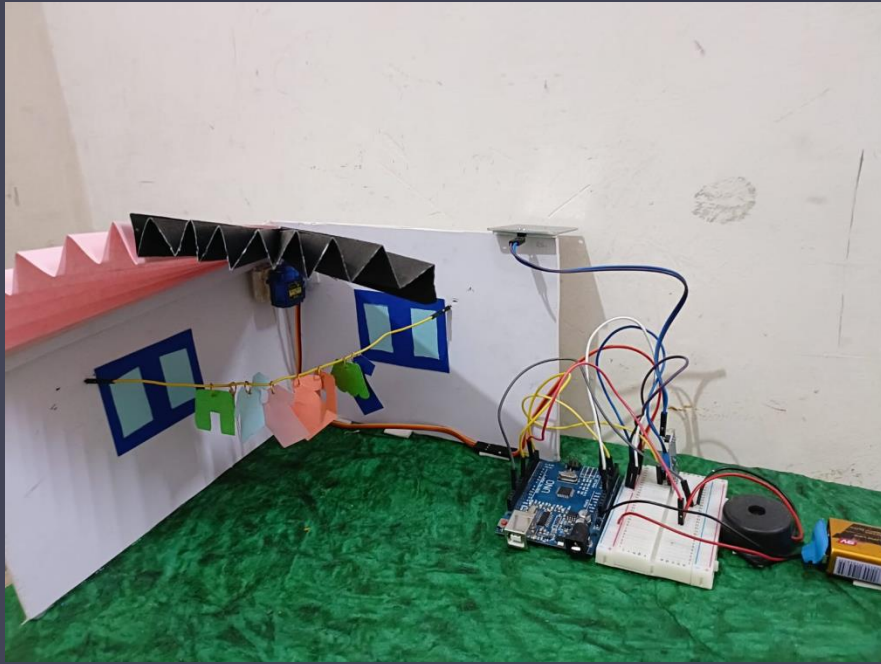
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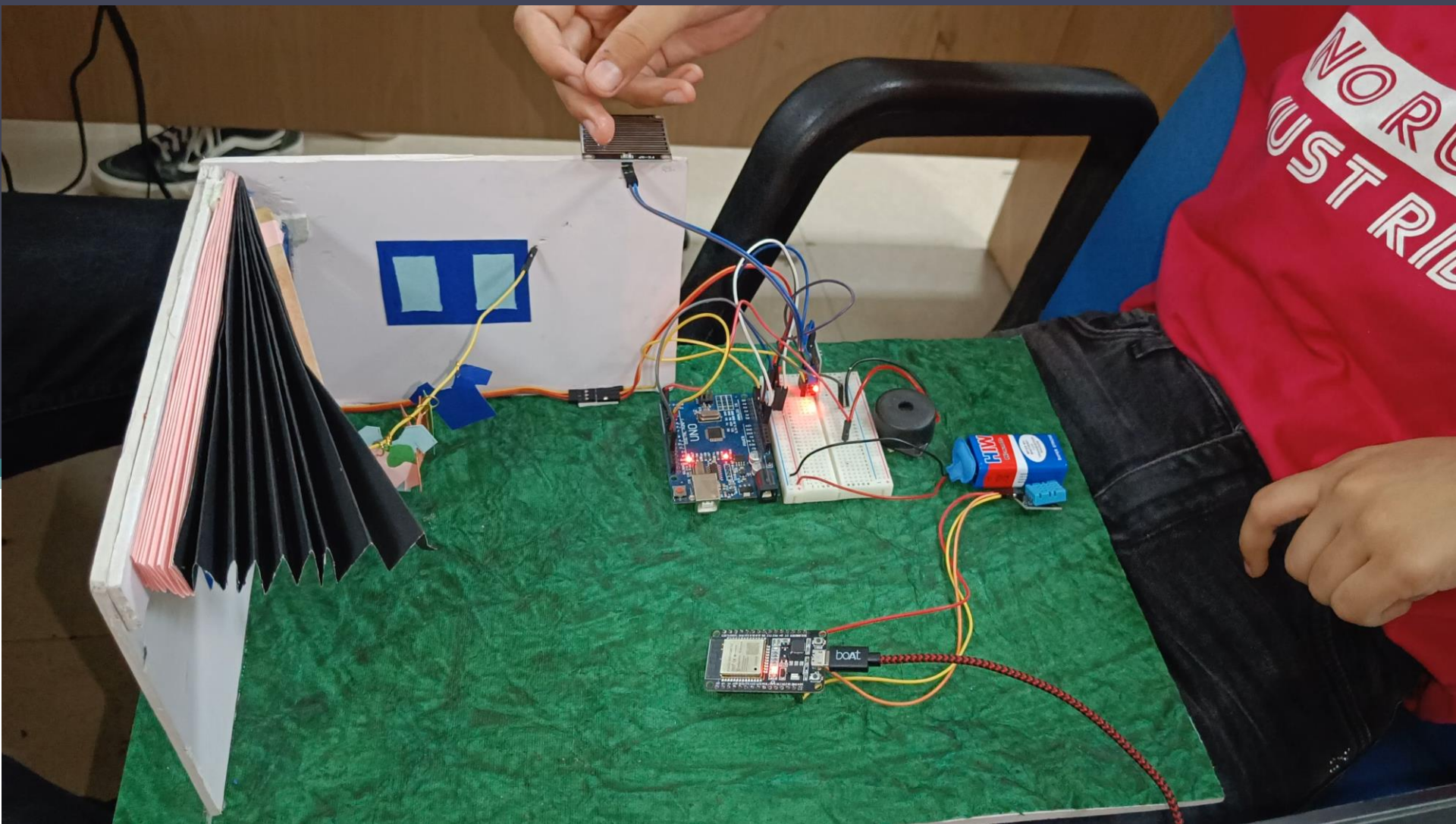


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IMAGES OF OUR PROJECT





ADVANTAGES of the project.

Here are some key advantages of using IoT (Arduino,ESP32) for a rain alarm and shutter system:

- Automation: An IoT-based system can automatically detect rain and close the shutters accordingly without any human intervention. This can be done using rain sensors to detect rain and motors to control the shutters.
- Customization: The system can be customized to your specific needs. For example, you can set different threshold levels for the rain sensor to trigger the shutter closing. You can also configure timers to automatically open/close the shutters at certain times.
- Low cost: Using Arduino and ESP microcontrollers along with basic sensors and motors allows you to build such a system at a low cost. These development boards and components are very affordable.
- Energy efficiency: An automated system will close the shutters only when needed, which can help reduce heating/cooling costs.

THANK YOU

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